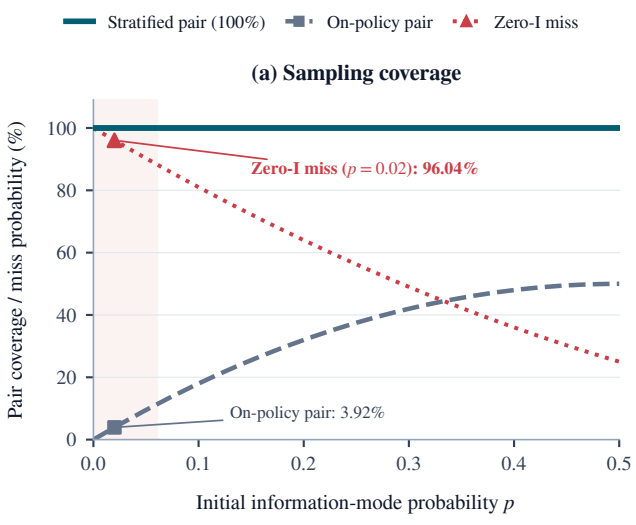


**(b) Gradient agreement (MC: 10,000)**

Objective $J(\theta_0)$ **58/125 = 0.4640**

Variance identity: LHS = RHS **0.0164736**

| **Estimator** | $g_1$ | $g_2$ | $g_3$ |
|---|---|---|---|
| Analytical | +0.03840 | +0.04480 | +0.00000 |
| **Stratified exact** | **+0.03840** | **+0.04480** | **+0.00000** |
| RLOO exact | +0.03840 | +0.04480 | +0.00000 |
| Autograd | +0.03840 | +0.04480 | +0.00000 |
| **Stratified MC** | **+0.03814** | **+0.04439** | **-0.00031** |
| RLOO MC | +0.03736 | +0.04529 | -0.00164 |